AJINKYA REPALE

Quantitative Systems Developer · Options Market Structure · Algorithmic Trading

+91 8779260512 · ajinkyarepale28@gmail.com · LinkedIn · GitHub

SUMMARY

Second-year Computer Engineering student at VIT Mumbai who builds production-grade quantitative trading systems from scratch. Independently developed a live multi-alpha options engine for Nifty using GEX, IV term structure, OI unwinding, and Vanna/Charm flows — concepts typically found at institutional trading desks. Combines deep market structure knowledge with clean, modular Python to bridge the gap between theory and real markets. Seeking internship opportunities in quant research, algorithmic trading, or trading technology.

PROJECTS

Nifty Alpha Engine — GEX + IV Term Structure + Vanna/Charm

2024 – 2025

Python · Black-Scholes · NSE Options Chain · Live Signal Engine · Kite Connect

- Built a **production-grade**, live options alpha engine for Nifty from scratch — scans the full NSE options chain every 5 minutes during market hours and generates confidence-scored trade signals.
- Implemented a complete Black-Scholes engine independently: IV solver using Brent's numerical method, all standard Greeks (delta, gamma, theta, vega), and second-order Greeks — Vanna (delta sensitivity to IV) and Charm (delta sensitivity to time decay).
- Alpha Source 1 — **GEX (Gamma Exposure):** Calculates net gamma exposure across all strikes to classify the market into regimes (Pinned / Trending / Strong Trend), identifies the GEX flip level as a dynamic support/resistance zone, and determines whether options buying has positive or negative expected value.
- Alpha Source 2 — **IV Term Structure:** Detects dislocations where weekly IV trades below monthly IV — a statistically significant mispricing signal that creates vega expansion opportunities independent of market direction.
- Alpha Source 3 — **OI Unwinding Detector:** Identifies strikes with >20% OI reduction intraday, detecting where option writers are covering positions and using it to time directional entries.
- Alpha Source 4 — **Vanna/Charm Flow Model:** Aggregates net vanna and charm exposure across all OI to predict the direction of market maker delta re-hedging flows — explaining directional drift that has no obvious catalyst.
- Signal aggregator scores all four sources on a weighted confidence scale (0–4+); only fires ENTER signals when GEX is negative (trending regime) AND at least one additional source aligns — enforcing strict signal quality control.
- Logs all signals to a structured JSONL file; backtester replays logs and calculates 7 performance metrics: win rate, reward/risk, expectancy, profit factor, Sharpe ratio, max drawdown, and max consecutive losses vs defined target ranges.

Gold London Session Liquidity Backtest — XAUUSD

2024

Python · Pandas · NumPy · Strategy Backtesting · ICT/SMC Concepts

GitHub: github.com/ajinkyarepale/Gold-London-Liquidity-Backtest

- Built a full Python backtesting engine to evaluate a London session liquidity sweep strategy on XAUUSD — programmatically detecting Asia session ranges, liquidity sweeps above/below key highs & lows, and Break-of-Structure (BOS) confirmation.
- Simulated realistic trade execution with dynamic entry, stop-loss, and take-profit logic; computed R-multiples, win rate, expectancy, max drawdown, and quarterly consistency — evaluated on risk-adjusted metrics, not raw win rate alone.
- Designed with an explicit anti-curve-fitting philosophy: out-of-sample validation across multiple market regimes before any live consideration.

TECHNICAL SKILLS

Languages & Tools: Python (Pandas, NumPy, SciPy), Git & GitHub, Kite Connect API

Quant / Math: Black-Scholes Pricing, IV Solving (Brent's Method), Greeks & Second-Order Greeks (Vanna, Charm), GEX Modelling, Backtesting, Sharpe Ratio, Drawdown Analysis

Options & Derivatives: IV Term Structure, Options Mispricing, GEX Regimes, OI Analysis, Vanna/Charm Flows, Market Maker Hedging Dynamics

Trading & Markets: Price Action, Support/Resistance, Candlestick Patterns, RSI, MACD, ICT/SMC (Liquidity Sweeps, BOS, Order Blocks, FVGs), Session Analysis

Markets Covered: Nifty 50 & Bank Nifty Options, NSE Equities, XAUUSD, EURUSD, GBPUSD, USDJPY

EDUCATION

Vidyalankar Institute of Technology, Mumbai

2024 – 2028

Bachelor of Engineering — Computer Engineering

- Relevant coursework: Data Structures & Algorithms, Discrete Mathematics, Object-Oriented Programming.
- All quantitative finance and trading systems work is fully self-directed and built independently alongside core curriculum.

ADDITIONAL

- Maintains a structured daily trading journal with weekly performance review and iterative strategy refinement across Nifty options and forex pairs.
- Deep interest in market microstructure, institutional order flow, and building software infrastructure for systematic trading.
- Available immediately for internships in quant research, algorithmic trading, options trading technology, or fintech development.